

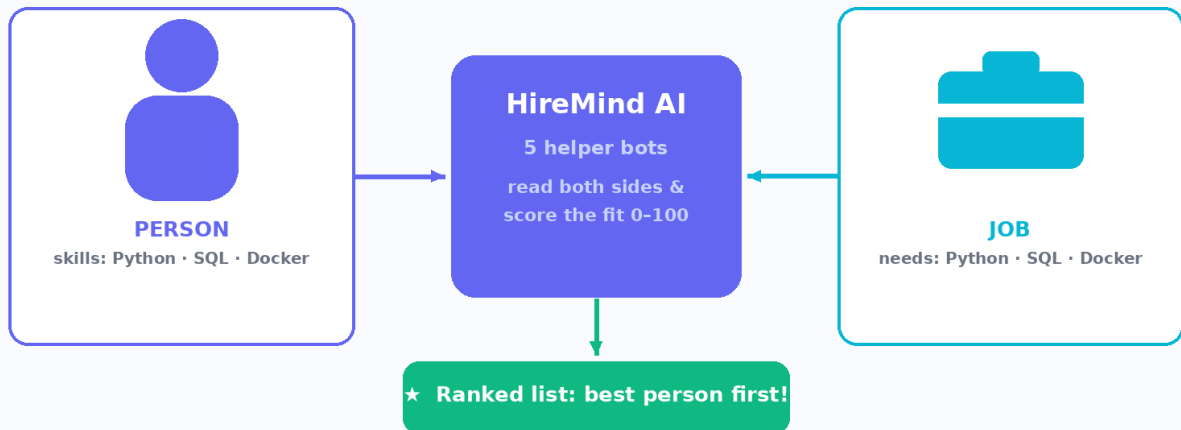
HireMind AI

The Robot Team that Matches People to Jobs

[Project Report](#) · [Hackathon Problem PS03](#) · [explained so simply a 10-year-old gets it](#)

The Big Idea: match the right person to the right job!

Like fitting puzzle pieces together — automatically, fairly, and in seconds.



A person's skills and a job's needs are like two puzzle pieces — HireMind AI measures how well they fit.

Team	[YOUR TEAM NAME] — [MEMBERS]
Hackathon	[HACKATHON NAME]
Category	NLP + LLM + Agentic AI
Date	20 Sept 2026

What's inside

- 1. The Problem — imagine a teacher with 300 homework sheets
- 2. Our Solution — 5 helper bots working like a relay race
- 3. The Score Pie — grading with fixed, visible rules
- 4. A Real Example — Priya vs the ML Engineer job
- 5. Why It Is Fair — two brains: AI reads, math decides
- 6. Try It in 3 Clicks + Who Wins? Everybody!
- 7. The Real App — actual screenshots
- 8. Demo Results — 10 resumes × 3 jobs, ranked
- 9. Tech Stack, What's Next & Honesty Box
- Appendix — the grown-up technical details for judges

Meet the Team: 5 Helper Bots (agents)

Each bot does ONE job, then passes the work to the next — like a relay race.



Output: ranked candidates + reasons for every score!

The five agents, one after another — the heart of the whole project.

1. The Problem (imagine this!)

Imagine a teacher who must check **300 homework sheets in one night**. That is a recruiter: one job ad can bring hundreds of resumes, and they can look at each for only about **7 seconds**.

So three bad things happen:

- Good people get missed just because they wrote a skill differently ("React.js" vs "React").
- Two recruiters can judge the same resume differently — not fair, not consistent.
- Nobody explains *why* someone was rejected.

Our mission: build a computer team that reads every resume carefully, scores it with clear public rules, and always explains why.

2. Our Solution: 5 Helper Bots

We built **HireMind AI**: five small programs called **agents** (think helper robots). Each does one job and passes the work on, like a relay race:

- **Reader Bot** reads a resume → tidy list of skills, years of practice, school.
- **Job Bot** reads the job ad → what the job **MUST** have vs what is extra-nice.
- **Score Bot** compares both lists → score 0–100 with fixed rules.
- **Gap Bot** finds missing skills → suggests what to learn next, like a game skill tree.
- **Writer Bot** writes a friendly summary for the recruiter.

How Score Bot grades — like a test with clear rules



The score is a pie with fixed slices — everyone can see exactly how points are earned.

3. A Real Example from Our Demo

We tested with 10 made-up resumes and 3 made-up job ads. Here is one real match, exactly as the bots computed it:

Example: does Priya fit the Senior ML Engineer job?

Job must-haves

- ✓ Python — she has it!
- ✓ Machine Learning — she has it!
- ✓ SQL — she has it!
- ✗ Docker — missing!
- ✗ PyTorch — missing!

(+2 more matched, not shown)

Priya's score

69

out of 100 = GOOD MATCH

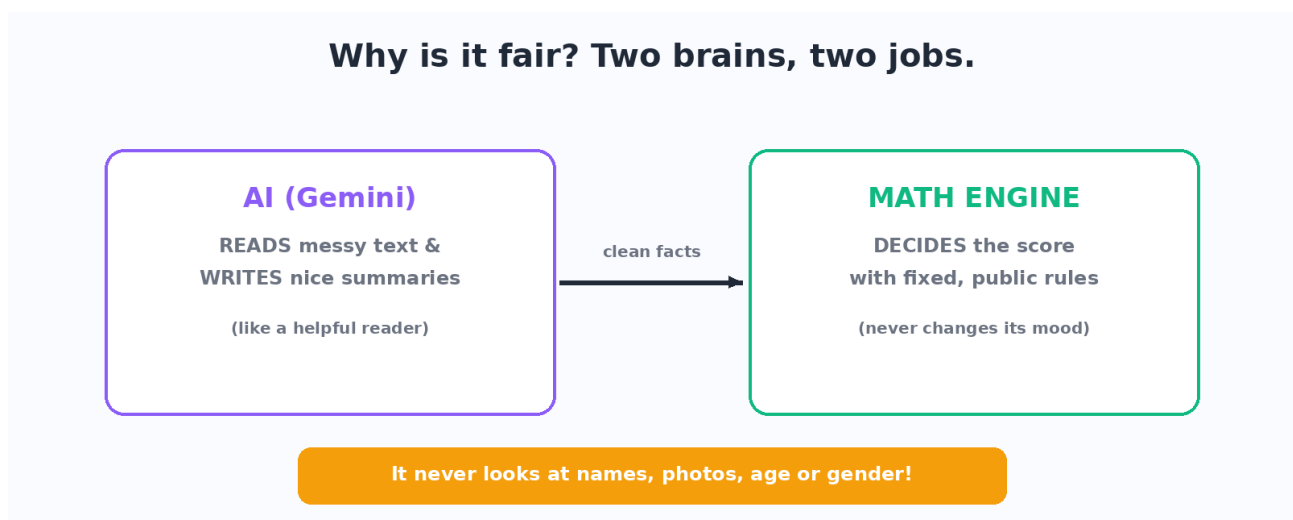
★ ★ ★ ★ ★

Gap Bot: learn Docker & PyTorch
→ strong candidate in ~1 month!

Priya has Python, ML and SQL but no Docker/PyTorch → 69/100 = Good match, plus a learning plan from Gap Bot.

See the magic? A “no” becomes a **to-do list**: learn Docker and PyTorch, and she becomes a strong candidate.

4. Why It Is Fair and Trustworthy



The AI reads and writes. The math decides. Nobody can argue with public, fixed rules.

- It **never looks** at names, photos, age or gender — only skills, practice and school.
- "React.js", "ReactJS" and "React" count as the **same skill** (dictionary of 130+ skills).
- Every score shows its **working**: ✓ matched skills, ✗ missing ones, points per slice.
- Even if the internet or the AI dies, the app still works — a built-in “no-AI” mode.

5. Try It in 3 Clicks — and Who Wins? Everybody!



The whole live demo takes under 30 seconds — perfect for judging.



Recruiters save hours and stay fair; candidates get reasons and a learning path.

6. The Real App (actual screenshots)

Everything above is a real working website: a React screen in front, a Python brain behind.

**HireMind AI**
Multi-Agent Resume Screening & Job Matching · PS03

Offline NLP mode — add a Gemini key for LLM agents

Settings

1 Resume Agent → 2 Job Agent → 3 Matching Agent → 4 Skill Gap Agent → 5 Recruiter Agent

1 · Resumes 102 · Job Descriptions 33 · Ranked Matches

Upload resumes (.txt / .pdf / .docx)

< Load demo dataset (10 resumes + 3 JDs)

10 resume(s) in store

Paste resume text

filename (optional), e.g. riya_resume.txt

Priya Verma
Data Engineer
priya@example.com

SKILLS
Python, SQL, Airflow, Spark...

EXPERIENCE

+ Add resume

Aarav Sharma

offline

Machine Learning Engineer · 5 yrs exp · B.TECH

✉ aarav.sharma@example.com

airflowawsdeep learningdockeretl
gitkubernetesmachine learning+9 more

Priya Nair

offline

Data Scientist · 4 yrs exp · M.SC

✉ priya.nair@example.com

ab testingdata analysisgitmachine learning
nlppandaspythonr+5 more

Rohan Mehta

offline

Frontend · 3.5 yrs exp · MBA

✉ rohan.mehta@example.com

cssgithtmljavascriptjestnext.js
reactredux+4 more

Sneha Kulkarni

offline

Full-Stack · 4 yrs exp · MBA

Vikram Singh

offline

Backend · 6 yrs exp · B.TECH

Ananya Iyer

offline

Devops · 5 yrs exp · B.E

Reader Bot finished: 10 resumes turned into neat cards with skills, years and school.

HireMind AI · PS03 Project Report (simple edition)

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Senior Machine Learning Engineer — Finlytics AI (Pune / Remote) Full-Stack Web Developer — UrbanKart (Bengaluru)

Data Analyst — RetailWave (Ahmedabad / Hybrid)

▶ Run Matching Pipeline 10 candidate(s) × 5 agents

Ranked 10 candidates for **Senior Machine Learning Engineer — Finlytics AI (Pune / Remote)** · Matching Agent: deterministic weighted scoring · Recruiter summaries by template engine

1 **Aarav Sharma**
Machine Learning Engineer · 5 yrs exp · B.TECH

Strong match

95

MUST-HAVE 100

NICE-TO-HAVE 67

EXPERIENCE 100

EDUCATION 100

KEYWORDS 100

2 **Priya Nair**
Data Scientist · 4 yrs exp · M.SC

Good match

69

MUST-HAVE 57

NICE-TO-HAVE 33

EXPERIENCE 100

EDUCATION 100

KEYWORDS 83

3 **Divya Reddy**
ML Engineer · 0.7 yrs exp · M.TECH

Partial match

59.2

MUST-HAVE 71

NICE-TO-HAVE 33

EXPERIENCE 10

EDUCATION 100

KEYWORDS 100

4 **Ananya Iyer**
Devops · 5 yrs exp · B.E

Partial match

53.4

MUST-HAVE 29

NICE-TO-HAVE 33

EXPERIENCE 100

EDUCATION 100

KEYWORDS 56

5 **Sneha Kulkarni**

Partial match

50.9

Select a candidate
Click any ranked card to open the explainable match report: score breakdown, skill gaps and recruiter summary.

Ranked list for “Senior ML Engineer”: best match on top, mini score bars on every card.

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50.9

95 **Aarav Sharma** Strong match

Machine Learning Engineer · 5 yrs exp · B.TECH · ranked #1 for “Senior Machine Learning Engineer — Finlytics AI (Pune / Remote)”

RECRUITER SUMMARY OFFLINE

Aarav Sharma is a machine learning engineer with 5 years of experience, scoring 95.0/100 (strong match) for Senior Machine Learning Engineer — Finlytics AI (Pune / Remote). Key strengths: deep learning, docker, machine learning, python. All must-have requirements are covered. Recommend moving to interview — strong alignment with the role.

MUST-HAVE COVERAGE

deep learning

docker

machine learning

python

pytorch

sql

tensorflow

NICE-TO-HAVE COVERAGE

aws

kubernetes

nlp

sagemaker

llm

spark

SCORE BREAKDOWN (DETERMINISTIC)

Must-have skills · weight 45% 100 — 7/7 matched

Click a person → the full WHY: green ✓ skills, red X gaps, recruiter note, every bot's work.

7. Demo Results (10 resumes × 3 jobs)

Job	#1 (best)	#2	#3
Senior ML Engineer	Aarav — 95 Strong	Priya — 69 Good	Divya — 59 Partial
Full-Stack Developer	Sneha — 100 Strong	Rohan — 81 Strong	Vikram — 66 Good
Data Analyst	Karan — 80.7 Strong	Priya — 78 Good	Aarav — 46.7 Partial

Common sense wins: the ML person tops the ML job, the web person tops the web job, the analyst tops the analyst job. A brand-new graduate scores lower on the *senior* job because practice years count — just like real life.

8. What We Built It With (tech stack)

Part	Tool	In kid words
Front screen	React + Vite	The colorful website you click
Back brain	Python + FastAPI	The kitchen where bots cook
AI reader/writer	Google Gemini 2.5 Flash	Smart helper for messy text (optional)
No-AI mode	Rules + skill dictionary	Backup brain so the demo never breaks
File readers	pypdf · python-docx	Opens PDF and Word resumes

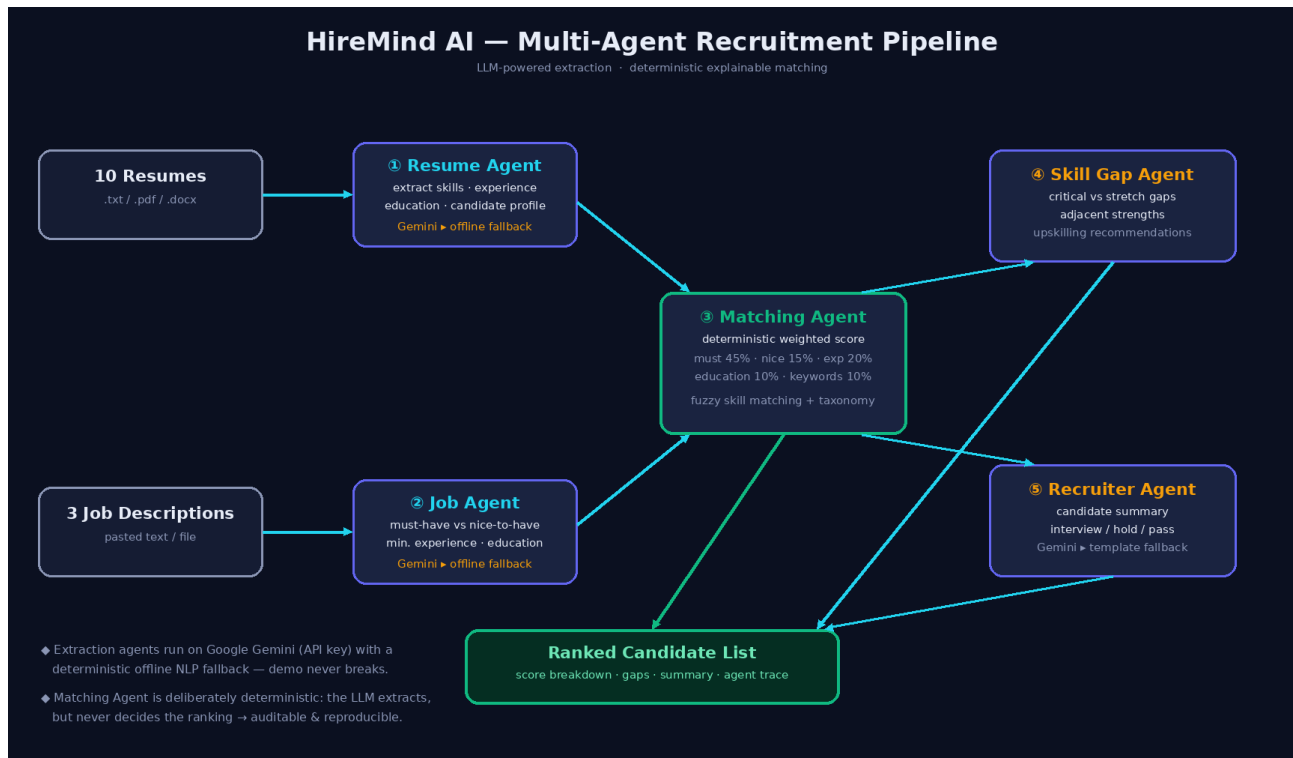
- Folders: **backend/** the 5 bots · **frontend/** React screens · **data/** 10 resumes + 3 jobs · **docs/** report, slides, pictures.

9. What's Next & Honesty Box

- Next: save results in a database, compare jobs side-by-side, export PDF lists for recruiters.
- Honesty: demo data is 100% made-up (no real people). Gemini is used only when a key is added. An AI coding assistant helped us type faster — the team chose every rule and checked every result.
- Judges: the grown-up details (weights, API, references) are in the appendix.

Appendix — Technical Details (for judges)

Scoring weights: must-have 45% · nice-to-have 15% · experience 20% · education 10% · keywords 10% (renormalised when a dimension is absent). Fuzzy skill match = alias dictionary + containment + difflib ratio ≥ 0.85 . Verdicts: ≥ 80 Strong, 60–79 Good, 40–59 Partial, <40 Weak. Deterministic $\Rightarrow$ reproducible.



Full technical architecture: Gemini/offline extraction feeds deterministic Matching, Skill Gap and Recruiter agents.

API	What it does
POST /api/resumes[/upload] · /api/jobs	Register documents → Agents 1 & 2
POST /api/demo/load	Load the 10 + 3 demo dataset
POST /api/match {job_id}	Agents 3–5 → ranked list + traces
POST /api/reextract	Re-run extraction after adding a Gemini key

References: [1] Gemini API — ai.google.dev · [2] FastAPI — fastapi.tiangolo.com · [3] React — react.dev · [4] Vite — vite.dev · [5] python-docx / pypdf docs · [6] Python difflib · [7] Kaggle & Hugging Face resume/JD datasets (surveyed; verify licenses before use).

Run it: `cd backend && pip install -r requirements.txt && uvicorn main:app --host 0.0.0.0 --port 8000` → open <http://localhost:8000> → Load demo dataset → Run Matching Pipeline.